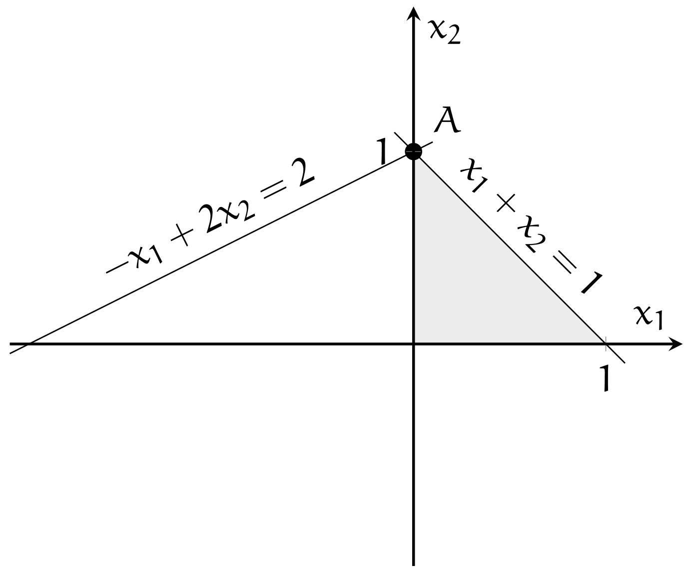
Consider the polyhedron defined by the following constraints:

$$\begin{aligned} x_1 + x_2 &\leq 1, \\ -x_1 + 2x_2 &\leq 2, \\ x_1, x_2 &\geq 0. \end{aligned}$$



In standard form, after introducing the two slack variables x_3 and x_4, we have

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ -x_1 + 2x_2 + x_4 &= 2, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The vertex A corresponds to $x_1 = 0$, $x_2 = 1$, $x_3 = 0$, $x_4 = 0$, and is degenerate. Consider the case where the variables x_1 and x_2 are in the basis. Calculate all basic directions, determine if they are feasible of not, and represent them on the picture.